

KALPESH SHARMA

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PROFESSIONAL SUMMARY

Automation-first Backend Engineer specializing in core platform infrastructure, integration frameworks, and scalable API ecosystems. Proficient in Python and distributed systems development with a strong grasp of computer science fundamentals. Experienced in designing high-performance middleware, building developer-centric SDK patterns, and integrating AI-driven automation tools to optimize the end-to-end software development lifecycle (SDLC) while eliminating manual overhead.

EDUCATION

Dr. B.R. Ambedkar National Institute of Technology, Jalandhar
Bachelor of Technology (B.Tech)
CGPA: 7.89 / 10.00

Jalandhar, Punjab
2022 – 2026

SKILLS & TECHNOLOGIES

Platform Architecture	Integration Frameworks, Core SDK Design, High-Performance Middleware, Standardized API Paths
Programming	Python, Go (Exposure), Java, C++, SQL (PostgreSQL, MySQL, T-SQL), Bash Scripting
Backend	API Ecosystem & RESTful API V2 Design, Microservices, Distributed Systems
Automation & DevOps	Automation-First Infrastructure, CI/CD Pipelines, GitHub Actions, Docker Containerization, Git
Testing & Observability	Unit/Integration/E2E Test Automation, Production Logging, Metrics Tracking, Performance Tuning
AI Tooling	GenAI Frameworks, LLM Coding Assistants (Copilot, Cursor), Automated Connectivity Workflows

EXPERIENCE

Backend Platform Engineering Intern — Raycom Technology Jun 2025 – Aug 2025

- Built and maintained core features of internal integration and API frameworks using Python, expanding data workflow ingestion speeds by 25%.
- Promoted a "Zero-KTLO" automation strategy by identifying manual process bottlenecks and designing scripted pipeline solutions.
- Engineered reusable validation components and high-leverage middleware paths to guarantee structural consistency for third-party connections.
- Leveraged AI-driven coding tools and co-pilots within production sprints to accelerate clean, highly testable backend script delivery.

Data Integration / Software Intern — Ascore Technologies Jun 2024 – Aug 2024

- Optimized transactional retrieval middleware across Microsoft SQL Server and distributed endpoints, shrinking system API latency by 30%.
- Constructed standardized interface logic to streamline database access layers, enabling external modules to connect reliably.
- Exercised strict test driven automation workflows, conducting end-to-end integration mapping across multiple staging environments.

Research Intern (Data Analysis) — IIT Jodhpur Jun 2024

- Designed automated preprocessing micro-utilities in Python to handle high-velocity algorithmic data tracking operations.
- Authored clean documentation profiles and execution maps to provide standardized setup frameworks for downstream engineering units.

PROJECTS

Sentinel – AI-Driven Integration & API Framework [\[GitHub\]](#)

- Architected a high-concurrency developer utility using FastAPI and OpenAI APIs to simplify complex semantic data processing queries.
- Engineered standardized API endpoints and structured routing paths backed by a FAISS vector database to ensure discoverable third-party integrations.
- Incorporated continuous performance monitoring and request logging systems to handle real-time input verification automatically.

Aether – Fault-Tolerant Distributed Task Orchestration Engine [\[GitHub\]](#) [\[Demo\]](#)

- Designed an automation-first scheduling framework in Python featuring asynchronous execution loops and state persistence mechanics.
- Standardized multi-node communications inside containerized Docker setups, providing a resilient platform architecture for background tasks.

Mimir – High-Throughput Decoupled Event System [\[GitHub\]](#) [\[Demo\]](#)

- Developed an open communication application using a decoupled React.js client interface integrated smoothly with an optimized Python server.
- Standardized data payload structures and lifecycle pipelines to process variable parameter inputs under low operational latency boundaries.